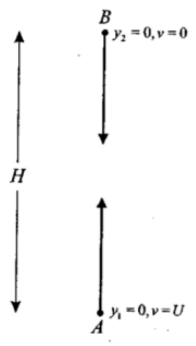
Mechanics WS 7

1. An object is projected downwards with initial velocity V. The air resistance at speed v has magnitude mkv, where k is a positive constant. Take downwards as the positive direction.
   a. Show that $\mathrm{t} = \frac{1}{\mathrm{k}} \ln\left(\frac{\mathrm{g}-\mathrm{kV}}{\mathrm{g}-\mathrm{kv}}\right)$
   b. Hence show that $\mathrm{v} = \frac{\mathrm{g}}{\mathrm{k}}\left(1 - \mathrm{e}^{-\mathrm{kt}}\right) + \mathrm{Ve}^{-\mathrm{kt}}$, and show that the terminal velocity is $\frac{\mathrm{g}}{\mathrm{k}}$.
   c. Integrate again to show that $\mathrm{x} = \frac{\mathrm{gt}}{\mathrm{k}} + \frac{\mathrm{kV}-\mathrm{g}}{\mathrm{k}^2}\left(1 - \mathrm{e}^{-\mathrm{kt}}\right)$.
   d. Suppose that the terminal velocity of this object is 20m/s, and that $\mathrm{g} = 10\mathrm{m/s}^2$. The object is thrown vertically downwards from a lookout at the top of a cliff at precisely the terminal velocity. At the same instant, a similar object is dropped from the same height. Show that the distance between the two falling objects after t seconds is $40(1 - \mathrm{e}^{-\frac{1}{2}\mathrm{t}})$
2. A particle P_1 of mass m is projected vertically upwards from a point A with initial velocity U. At the same instant, a second particle P_2, also of mass m, is dropped from a point B directly above A. The distance H between A and B is equal to the maximum height that P_1 would reach were it not to collide with P_2. As the particles P_1 and P_2 move, they each experience air resistance of magnitude mkv^2, where k is a positive constant and v is velocity. At the instant the particles collide, P_2 has reached 50% of its terminal velocity V. Let y_1 be the distance of P_1 above A, and y_2 the distance of P_2 below B.



   a. Show that $V = \sqrt{\frac{g}{k}}$.
   b. Show that $y_1 = \frac{1}{2k} \ln\left(\frac{g+kU^2}{g+kv^2}\right)$, where v is the velocity of P_1.
   c. Hence show that $H = \frac{1}{2k} \ln\left(1 + \frac{U^2}{V^2}\right)$.
   d. Assuming that $y_2 = \frac{1}{2k} \ln\left|\frac{g}{g-kv^2}\right|$, show that at the instant the particles collide, $y_2 = \frac{1}{2k} \ln\frac{4}{3}$.
   e. Deduce that the speed of P_1 at the instant the particles collide is $\frac{V}{\sqrt{3}}$.
3. On a certain day, the depth of water in a harbour at high tide is 11 metres. At low tide $6\frac{1}{4}$ hours later, the depth of water is 7 metres. If high tie is due at 2: 50 AM, what is the earliest time after midday that a ship requiring a depth of at least 10 metres of water can enter the harbour.
4. Tidal flow in a harbour is assumed to be in Simple Harmonic Motion and water depth x metres at time t hours is given by $x = 20 + A\cos(nt + \alpha)$ where A, n, and α are positive constants. The depth of water is 12m at low tide and 28m at high tide which occurs 7 hours later.
   a. Evaluate A and n
   b. On a day when low tide occurs at 2: 00am, find the first period of time during which the water level is greater than 22m.
5. Between 5am and 5pm on 3rd of March 2009, the height, h, of the tide in a harbour was given by $h = 1 + 0.7\sin\frac{\pi}{6}t$ for $0 \le t \le 12$, where h is in metres and t is in hours, with $t = 0$ at 5am.
   a. What is the period of the function h?
   b. What was the value of h at low tide, and at what time did low tide occur?
   c. A ship is able to enter the harbour only if the height of the tide is at least 1.35m. Find all times between 5am and 5pm on

3rd of March 2009 during which the ship was able to enter the harbour.

6. A piece of cork moves vertically in Simple Harmonic Motion on the surface of the water as waves pass under it. Its velocity v m/s is given by $v^2 = -x^2 + 7x - 12$, where x is the cork's vertical displacement in metres.
   a. Using $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$, find the acceleration of the cork in terms of x
   b. What is the centre of motion?
   c. Find the period of oscillation
7. A particle moves in a straight line such that its displacement x centimetres at any time t seconds is given by $x = \cos 3t + \sin 3t$
   a. Prove that the motion is Simple Harmonic Motion
   b. State the period of the motion
   c. Find the initial position of the particle
   d. Find the velocity of the particle when it first returns to its initial position. (You may use your results from Part b.)
8. The deck of a ship was 1.4 m below the level of a wharf at low tide and 0.6 m above wharf level at high tide. Low tide was at 8:24 am and high tide at 2:40 pm. If tide's motion is simple harmonic, find the first time after low tide that the deck was level with the wharf.
9. The acceleration of a particle is $(2x - 5)$ m/s^2, where x is the distance in metres from the origin.
   a. Find an expression for the velocity of this particle in terms of x, given that the particle is at rest one metre to the left of the origin initially.
   b. Describe the motion
10. The acceleration of a body moving along a straight line is given by: $a = -24e^{-4x}$ where x is the displacement from the origin after t seconds. Initially the particle is at the origin with velocity of $2\sqrt{3}$ m/s.
    a. Prove that the velocity, v, of the body in terms of x is given by $v = 2\sqrt{3}e^{-2x}$
    b. Find an expression for t in terms of x.
    c. Hence show that $x = \frac{1}{2}\ln(1 + 4\sqrt{3}t)$.
    d. How long does it take for the body to reach a point 5 m to the right of the origin? Leave your answer in exact form.
11. A particle of unit mass moves in a straight line. It is placed at the origin on the x-axis and is then released from rest. When at position x its acceleration is given by $-9x + \frac{5}{(2-x)^2}$. Prove that the particle ultimately moves between two points on the x-axis and find these points.
12. When a particle moving in a straight line has displacement x metres from a fixed point O, its acceleration in metres per second is $\sqrt{3x + 4}$
    a. Show that $v^2 = \frac{4}{9}(3x + 4)^{\frac{3}{2}} + c$, where v is the velocity of the particle in metres per second, and c is a constant
    b. Given that the particle starts from rest at O, find c
    c. Explain why the motion of the particle is always in the positive direction

Answers:

1. d. 40m
2.
3. 1:15 PM
4. a. $A = 8, n = \frac{\pi}{7}$; b. between 6:04am and 11:56am
5. a. $T = 12$ hours; b. 2pm at $h = 0.3$m; c. 6am to 4pm
6. a. $\ddot{x} = -x + \frac{7}{2}$; b. $x = \frac{7}{2}$; c. $period = 2\pi\ seconds$
7. b. $Period = \frac{2\pi}{3}$; c. $x = 1\ cm\ to\ the\ right$; d. $\dot{x} = -3\ cm/s$
8. 12:21 pm
9. a. $v = -\sqrt{2x^2 - 10}$; b. starts from $x = -1$ from rest, moves to the left with increasing speed, never stopping again and changing direction
10. b. $t = \frac{e^{2x}-1}{4\sqrt{3}}$; d. $t = \frac{e^{10}-1}{4\sqrt{3}}$ seconds
11. Proving
12. $c = -\frac{32}{9}$